

Lecture One

Have to internalise theories before using them - group theories from the start.

A subsection/topic is one page of lecture notes - three a week. There are then handouts with more examples and

I forgot how much this guy yaps - don't do daily games before his lectures, do them during >:)

IT HAS BEEN TWENTY MINUTES WITH NO CONTENT AHHHHHHHHHH

Nevermind love this guy, waggle waggle

Just realised Dan does way more work when it's his supervisor doing the lecture; will

Assessment Weighting: Portfolio (at-home/in-class assessments 1:3 split) 40% / Final Exam 60%

1.1. Introduction to Permutations

Permutation: re-arranging a set; a bijection (injective & surjective, one-to-one & onto) Just a complete unique m

The set (group) of all permutations of a set, X , is $\text{Sym}(X)$. If X is just $\{1, 2, 3, \dots, n\}$, we write S_n by convention in

The modulus of these sets (S_n), $|S_n|$, is $n!$ - initially n choices, but then just $n-1$ choices moving an element do

For finite X , $|\text{Sym}(X)| = |X|!$ - proven inductively like on the left, for the previous argument.

Considering a specific example of the former, S_7 , for example a permutation sigma within this set (see to the

These permutations always decompose into cycles, which can simplify our notation a bit: $(1)(2\ 4\ 3\ 6\ 7)(5)$ - the

By convention, we exclude one-cycles, like the aforementioned (1) or (5) - single element cycles.

Hence for our example, sigma would be $(2\ 4\ 3\ 6\ 7)$ in S_7 and excluded numbers are "fixed" (one-cycles).

Whenever he says six-seven I get naam flashbacks

Another more complex example (see left) exemplifies the usefulness of the new notation

NOTE: within permutations it's left $\rightarrow$ right, but when they're written next to each other they operate BETW

NOTATION NOTE: We write sigma x instead of sigma(x) to separate it from the bijection notation.

FORMAL PERMUTATION DEFINITION: Let $a_1, a_2, \dots, a_r$ be distinct elements of a set X . Then sigma = $(a_1\ a_2\ \dots\ a_r)$

These notes are gonna be annoying to sort, he's so all over the place grr

Pairwise Disjoint Definition: A collection of cycles in $\text{Sym}(X)$ is disjoint (NOTHING IN COMMON) if no element in X

For example, cycles $(1\ 2\ 3)$ and $(4\ 7\ 5)$ are disjoint, but not $(1\ 2\ 3)$ and $(4\ 7\ 5)$

Uh oh controversial remark (APPARENTLY VERY IMPORTANT): $(1\ 2\ 3)$ and $(3\ 1\ 2)$ are identical. YEAH DUH NOT V

1.2. Products of Permutations

Definition: Let sigma and rho be permutations of a set X . The product sigma rho means (see right). Essentially, v

This definition is by convention which conflicts between academics - look into this (!!)

"it's definitely easier to see more do it because I'll waffle" - yeah we know bucko

oh it's just like sigma(rho(1 2 3 4 5 ... N)) where N is from S_N , work within cycles too to make it easier

like sigma = $(1\ 2\ 3)$ and rho = $(1\ 5\ 3\ 4)$ in S_6 , given sigma rho, becomes: sigma rho 1 = sigma 5 = 5 sigma rho 5

Then keep going for the next cycle(s), now starting at 2... sigma rho 2 = 3 sigma rho 3 = 4 sigma rho 4 = 2 sigma rho

Then the last one is (6), fixed, so the final answer is: sigma rho = $(1\ 5)(2\ 3\ 4)(6)$.

Like composing functions - light work.

ew I just realised he writes tau really weird

Another example, sigma = $(1\ 2)$, rho = $(2\ 3)$, tau = $(3\ 5)$... sigma rho tau 1 = $(1\ 2)(2\ 3)(3\ 5)$ 1 = 2 - feeding the

Hence $(1\ 2\ 3\ 5)(4)$.

YET ANOTHER EXAMPLE: $(3\ 5)(2\ 4\ 6\ 8)(2\ 3\ 4) = ?(1)(2\ 5\ 3\ 6\ 8)(4)(7)$ - directly found by plugging in numbers in

Is kinda like simplifying permutations?

YIPPEEEEE IT'S OVER >:))))))

I'm stuck Stepbro!!
Lecture 2

"let

FREAKIER Proposition: Products of permutations are associative just composition of functions is.
IDENTITY permutation, e (or 1, Id, 0, I, among others) in $\text{Sym}(X)$ is the bijection that sends x to x for all x in X .
SUPER MEGA FREAKY Proposition: For all sigma and rho in $\text{Sym}(X)$:1. sigma $\circ$ e = sigma = sigma (identity element)
UBER SUPER MEGA FREAKY Proposition: Let X be a finite set. Every permutation in $\text{Sym}(X)$ can be written as a product of disjoint cycles.
DIDDY Proposition: Suppose sigma in $\text{SYmm}(X)$. We know we can write it as a product of disjoint cycles: sigma = $\gamma_1 \gamma_2 \dots \gamma_k$. The order of g in $\text{Sym}(X)$ is the smallest n in $\text{Sym}(X)$ that ... (idk I dozed off, probably something to do with in

A group is a set G with an operation $*$ on G such that:

1. Closure: for any two elements g, h , in the group, $g * h$ is also in G .
2. Associativity: $(g * h) * i = g * (h * i)$
3. Identity: there is an element e in G such that $g * e = g = e * g$
4. Inverse: for every g in G , there is an element g^{-1} in G such that $g * g^{-1} = e = g^{-1} * g$

Notice this allows groups to be written as permutations! Yippee!

if G & H are two groups then their direct product/cartesian product is a group denoted $G \times H = \{(g,h): g \in G, h \in H\}$

"nothing exciting now" yeah no shizzle my drizzle

Practical time bishes

$$\begin{array}{ccccccc}
 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 1.4.1.A & & & & & & \\
 4 & 7 & 2 & 1 & 5 & 6 & 3
 \end{array}
 \rightarrow (1\ 4)(2\ 7\ 3)(5)(6)$$